

Convolutional Neural Networks and Feature Maps

Research Internship Report — May 2026

Abstract

This report is my attempt to understand Convolutional Neural Networks (CNNs) and why they have become such an important topic in computer vision. I came across these ideas during my research internship and honestly found them very interesting once I got the basic intuition behind them. The report covers the basic structure of a CNN and explains how different layers work together to process images. It also discusses why CNNs are preferred over traditional fully connected neural networks for image classification and how they are able to automatically learn useful patterns from images.

Another part that I found interesting was feature map visualization. Looking at the intermediate outputs made it easier to understand what the network was actually learning at different stages. The report also briefly describes the training process used for the model and some of the concepts involved while building and visualizing it.

Keywords: CNN, CIFAR-10, convolution layer, feature maps, image classification.

1. Introduction

When I first started learning about image classification, one thing that became clear was that images are very different from normal numerical data. A fully connected neural network can be used for image classification, but it does not make good use of the spatial relationships present between pixels. This is where Convolutional Neural Networks become useful.

CNNs are designed to automatically learn patterns from images and have become one of the most widely used models in computer vision. Their ability to detect features at different levels makes them suitable for a large variety of image-related tasks.

2. CNN Architecture

In this assignment, a CNN model was implemented using two convolution blocks. Each block consisted of a convolution layer followed by Batch Normalization, ReLU activation and Max Pooling. The first block mainly extracted simple features, while the second block learned more complex patterns. After feature extraction, the output was flattened and passed through a fully connected layer to classify images from the CIFAR-10 dataset.

3. Feature Maps

To understand what the network learns internally, feature maps from different convolution blocks were visualized. The first convolution block mainly captured simple patterns such as edges, intensity changes and textures. The second block produced more abstract representations that contained richer information useful for classification. Looking at these feature maps helped me understand how information gradually changes as it moves deeper inside the network.

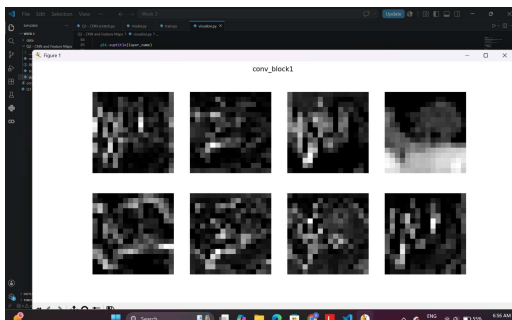


Figure 1: Feature maps obtained from the first convolution block.

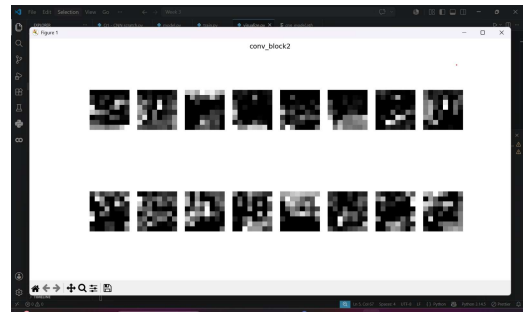


Figure 2: Feature maps obtained from the second convolution block.

4. Training Process

The model was trained using the Adam optimizer and Cross-Entropy Loss. During training, forward propagation and back-propagation were repeatedly performed to update the network parameters. Forward hooks were also used to capture intermediate activations for visualization. As training progressed over multiple epochs, the network gradually learned useful representations from the images and improved its predictions.

5. Conclusion

I think this assignment provided a good introduction to how Convolutional Neural Networks work internally. It helped me understand that image classification is not just about feeding pixels into a network, but about gradually extracting meaningful features through multiple layers. Visualizing feature maps also made it easier to connect the theory with what is actually happening inside the model.

It made the idea of CNNs and image processing feel much less mysterious to me and gave me a better understanding of how modern image classification models work.

References

1. PyTorch Documentation.
2. CIFAR-10 Dataset Documentation.